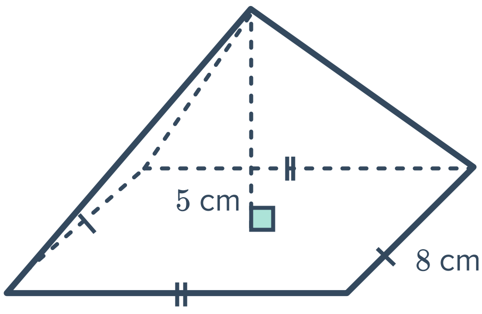
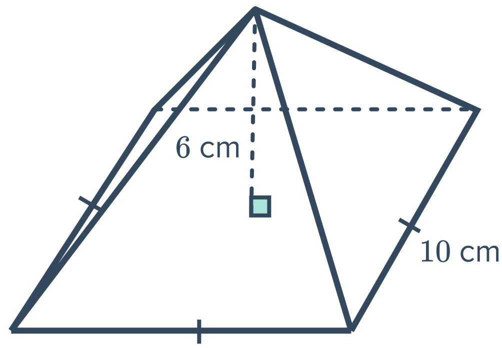
Volume of pyramids and cones

1 Find the volume of the following pyramids, correct to two decimal places if necessary:

a



b



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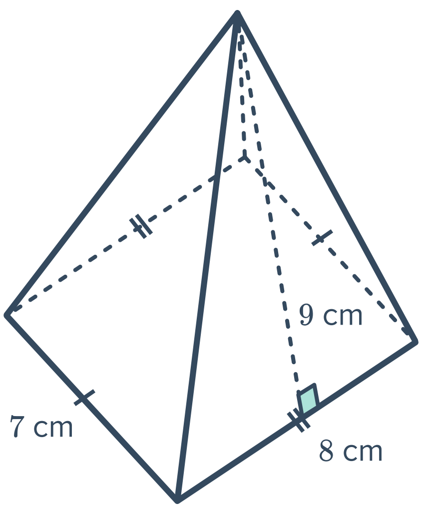
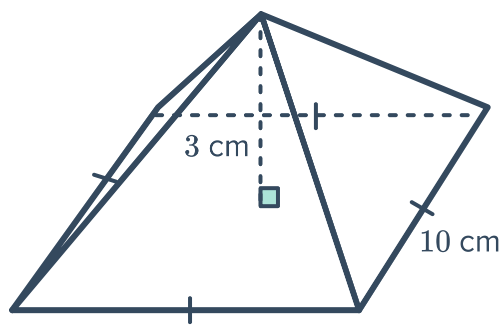
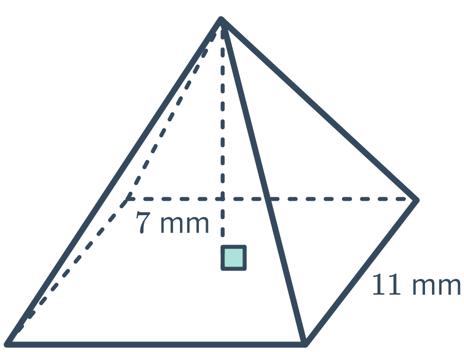


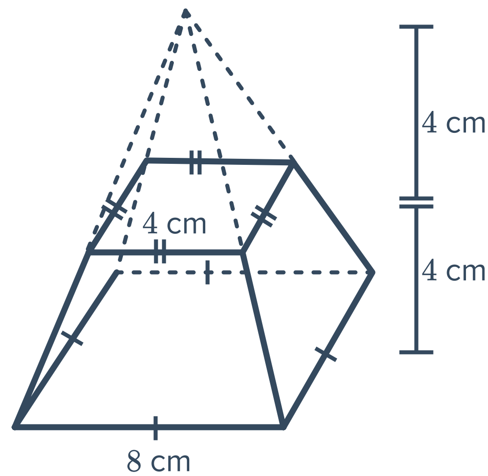
d



e



2 A small square pyramid of height $4\ \mathrm{cm}$ was removed from the top of a large square pyramid of height $8\ \mathrm{cm}$ forming the solid shown. Find the exact volume of the solid.



3 Suppose one particular pyramid in Egypt has a square base of side length $231\ \mathrm{m}$ and with a perpendicular height of $144\ \mathrm{m}$. Find the volume of the pyramid, correct to the nearest m^3.

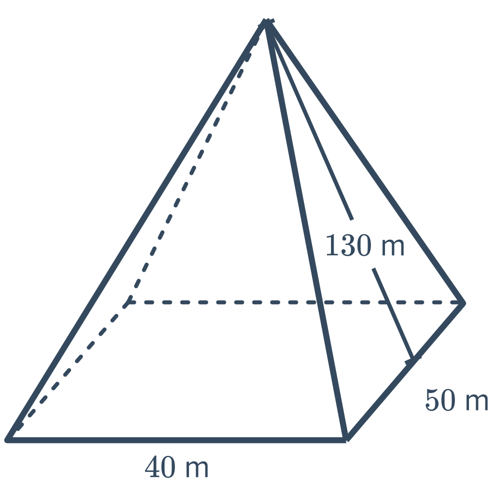
4 A rectangular pyramid has a volume of $288\ \mathrm{cm}^3$. The base has a width of $12\ \mathrm{cm}$ and length $6\ \mathrm{cm}$. Find the height of the pyramid.

5 Consider the following right pyramids:

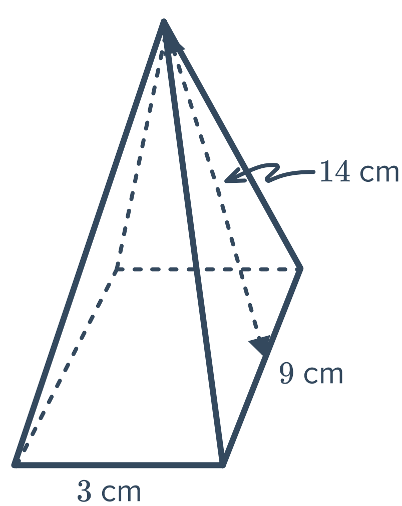
a Find the perpendicular height of the pyramid, correct to two decimal places.

b Hence, find the volume of the pyramid, correct to two decimal places.

a



b



6 Some very famous right pyramids are the Egyptian Pyramids.
The diagram shows the Great Pyramid. It has a square base of side length 230 m and a vertical height of 146 m. Find the volume of the Great Pyramid, correct to the nearest tenth.

7 Find the volume of the following right pyramids:

a A right pyramid with a square base of 24 mm and a perpendicular height of 24 mm.

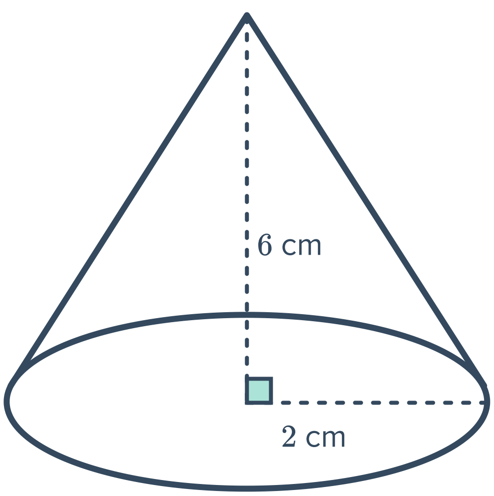
b A right pyramid with a square base of 21 mm and a vertical height of 27 mm.

8 A right square pyramid has a base of length 12 mm and a volume 1248 mm^3. Find the height of the pyramid.

9 A right square pyramid has a height of 24 cm and a volume 2592 cm^3. Find the base length of the pyramid.

10 Find the volume of the following cones, correct to two decimal places:

a



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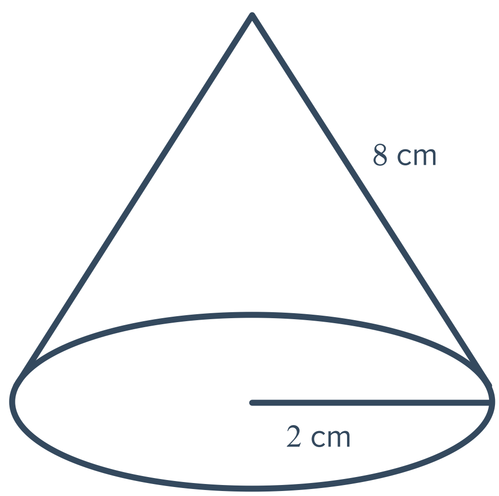

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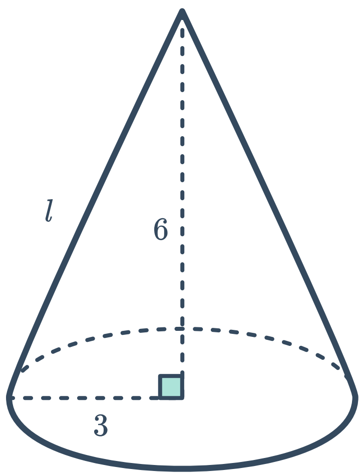
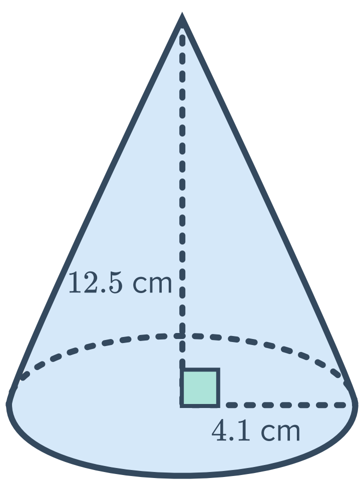


11 Find the volume of the following cones, correct to one decimal place:

a



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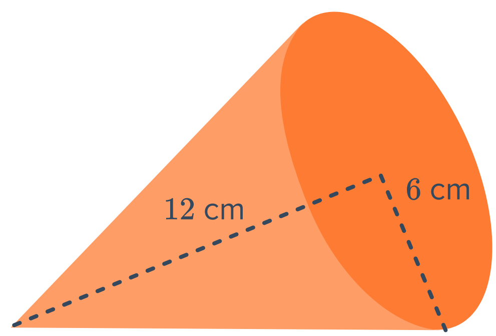

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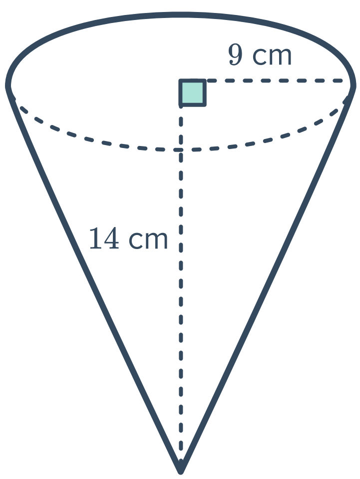
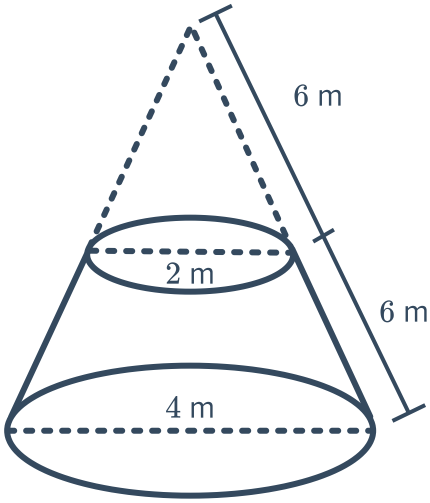


12 The podium was formed by sawing off the t[redacted] a cone. Find the volume of the following podiums, correct to two decimal places:

a



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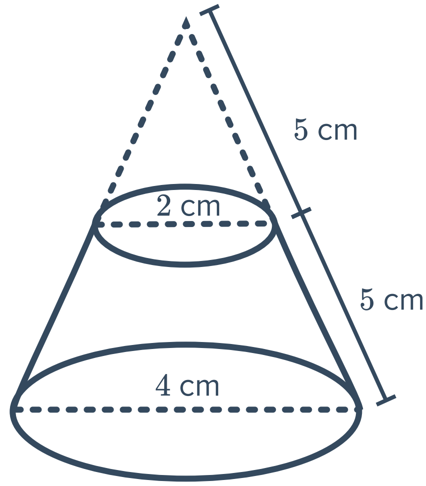


13 Find the volume of the following cones, correct to two decimal places:

a A cone with radius of 8 cm and vertical height of 13 cm.

b A cone with radius of 60 m and a vertical height of 27.54 m.

14 Find the height of the following cones:

a A cone with volume 603.19 m^3 and radius 6 m. Round your answer to one decimal place.

b A cone with volume $26\,640.97 \text{ cm}^3$ and radius 31.9 cm. Round your answer to the nearest cm.

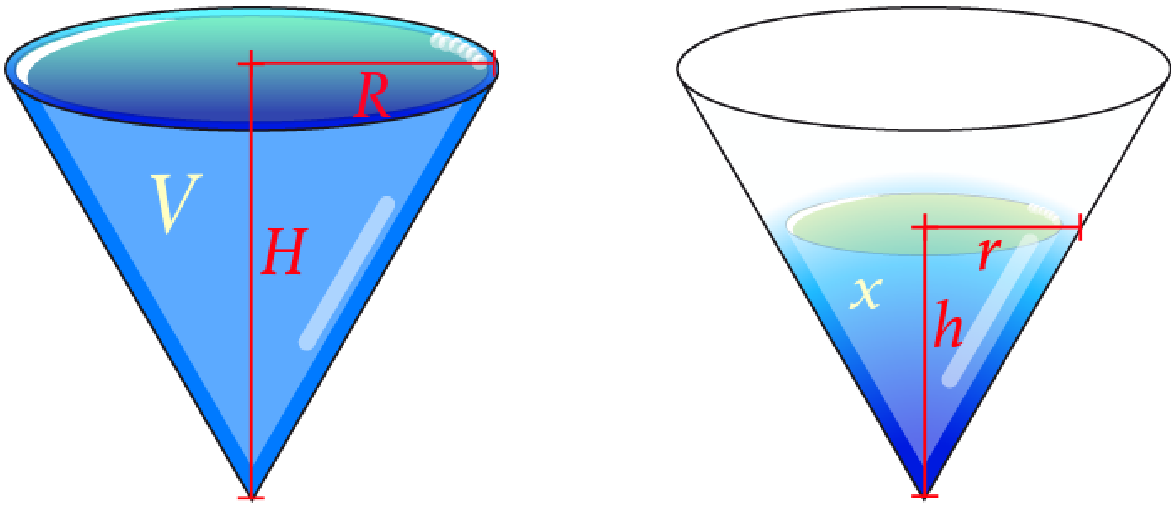
15 Find the radius of the following cones:

a A cone with volume of $12\,441.02 \text{ cm}^3$ and a height of 30 cm. Round your answer to one decimal place.

b A cone that has volume 1273.39 mm^3 and height 19 mm. Round your answer to the nearest mm.

c A cone with volume of 196 mm^3 and the length of the height and radius are equal. Round your answer to two decimal places.

16 A water vessel is produced in the shape of a cone of radius R and maximum volume V.
From vertex to base, the vessel has a height of H.
A volume x of water is poured into the vessel, filling up from the vertex towards the base.
The radius of the cone of water within the vessel has radius r and height h.

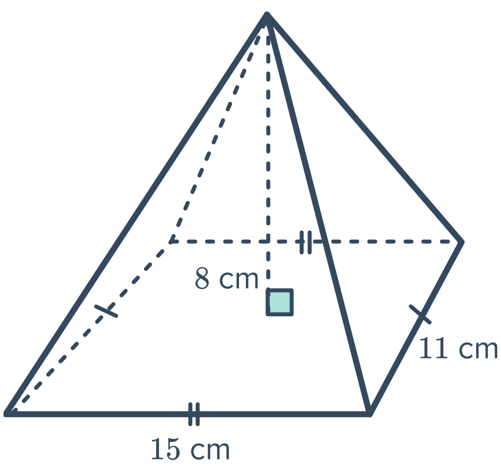


Write an expression for the height of the water, h, in terms of the height of the vessel H and the volumes V and x.

Additional questions

17 Find the volume of the following solids, rounding to two decimal places if necessary:

a



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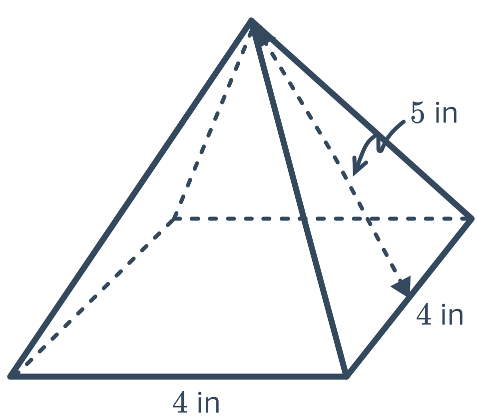

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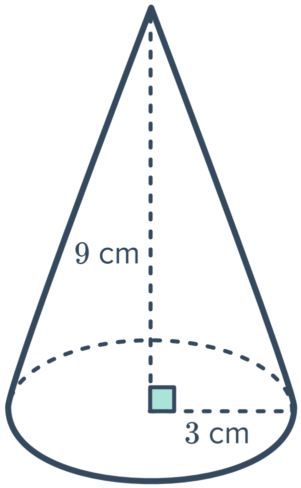
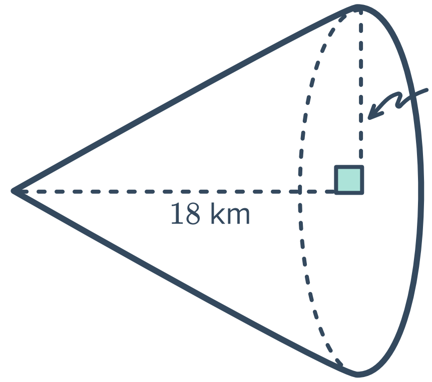
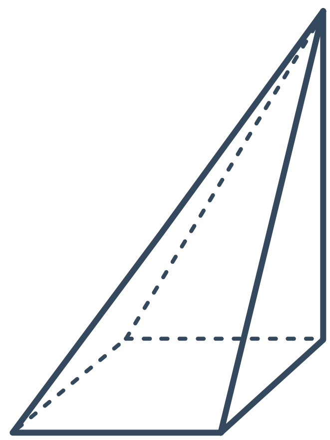




18 The volume of a right square pyramid is 55 in^2. If the pyramid is slanted by 60°, find its volume.

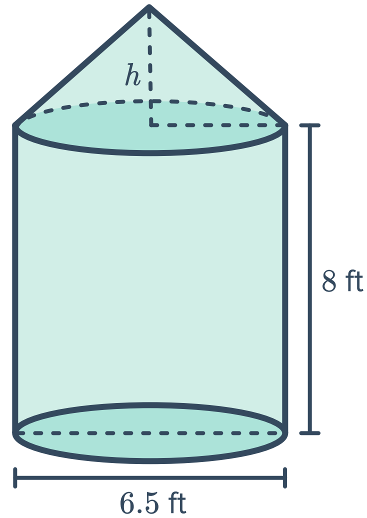


19 The radius and the height of a cone are equal, and the volume of the cone is 72π. Find the radius of the cone.

20 The volume of a pyramid is 60 m^3 with a height of 4 m. Find the area of the base.

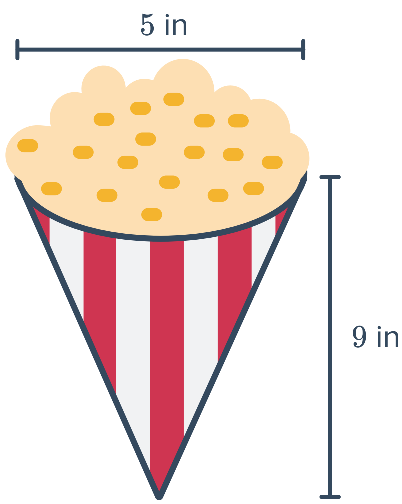
21 The Florida Agricultural Museum plans to bring in a replica grain silo for a new exhibit. The volume of the conical portion of the silo is $\frac{169}{24}\pi \text{ ft}^3$.

The museum's doorway has a height clearance of 12 ft . Will the model grain silo fit through the door? Explain.

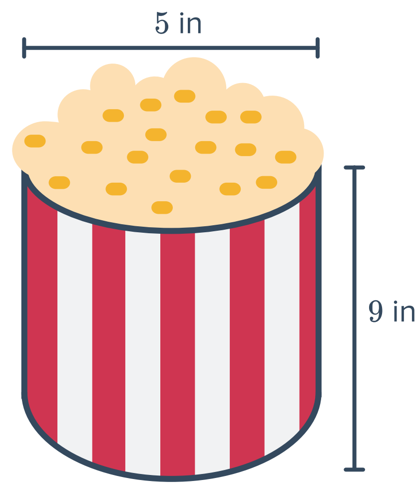


22 A vendor at the Sumter County Fair sells kettlecorn in two containers.

Option 1 sells for $\$3.50$:



Option 2 sells for $\$10.50$:



If you have $\$12.00$ to spend, what would you order to get the maximum amount of kettlecorn? Explain.